

Spatially Asynchronous Latent Generation in Irregular Image Inpainting: Limits of Geometry-Guided Phase and Content Control

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Abstract—Traditional generative sampling typically applies a uniform progress schedule across all spatial locations. This paper investigates spatially asynchronous latent generation in irregular image inpainting. We use a continuous phase to represent the local generation progress of different latent locations, comparing globally synchronous, geometry-guided, and content-controlled strategies under the same phase progression budget and number of main-model evaluations. Synthetic positive controls show that when data has a clear content-dependent formation order, a content-phase closed loop can learn this process. On four types of irregular masks from Places365, the geometry-guided phase reduces the unknown region latent MSE from 0.9711 to 0.9295 and the boundary MSE from 0.6196 to 0.4724. The improved Content-v2 outperforms the relative reliability baseline and cross-image Shuffle by 0.73% and 0.25%, respectively, but neither reaches the pre-specified practical effect thresholds of 2% and 1%. Furthermore, no reliable endogenous spatial precedence differences were observed in free generation trajectories. The results indicate that the geometry-guided phase can serve as an effective spatial generation prior, whereas current low-level latent content is not yet sufficient to be the primary basis for local progress control.

Index Terms—Image inpainting, spatially asynchronous generation, local progress, phase field, content control

I. INTRODUCTION

Diffusion models [1], [2] and Flow Matching [3] typically employ a uniform time for the entire image in a single sampling update. However, image inpainting exhibits obvious spatial non-uniformity [4]: locations near the known regions can directly utilize real boundaries and textures, whereas locations deep within unknown regions generally have higher uncertainty. Therefore, different locations may not need to complete generation synchronously.

Pixel-wise or region-wise time scheduling is not a new proposition. CVDm learns condition-dependent pixel-wise diffusion schedules [5]; RAD and AsyncDSB apply asynchronous processes to image inpainting [6], [7]; AsyncDM and Patch Forcing further utilize attention or local difficulty to adjust spatial generation progress [8], [9]. Related research has demonstrated that spatial locations can have varying generation speeds [10], [11]. Therefore, this paper no longer focuses on the necessity of local time, but investigates a more specific question: under the same phase progression

budget and number of main-model evaluations, should the local progression order in irregular inpainting be primarily determined by mask geometry, or can it be further optimized by the current generated content?

To this end, we introduce a continuous phase to represent the current generation progress of each latent location. The external mask always indicates the real known regions, while the phase only describes how far the generation process has progressed, without implying that the location has been generated correctly. Based on this representation, we compare globally synchronous, geometry-guided, and content-aware controls, and further design Content-v2 to decouple content control into local reliability and target utility for subsequent unknown regions. Experiments show that spatial asynchrony itself can stably improve inpainting, but content signals currently do not surpass simple and stable geometric priors.

II. SPATIAL ASYNCHRONOUS PHASE CONTROL

A continuous phase M is used to record the local generation progress of each latent location, remaining monotonic, bounded, and eventually completing during the sampling process. All strategies adopt the same total phase progression, initial noise, and number of main generative model calls, so the main difference lies in which locations are prioritized for progression.

Global advances all unknown locations synchronously; **Geometry** forms a smooth propagation based solely on mask geometry, spatial relationships to known regions, and the current phase interface; **Content** further reads the current latent in the same framework, aiming to adjust progression locations based on the generated content.

Content-v2 builds upon this by dividing content information into two categories: one estimates whether the generation state of the current location is stable, and the other estimates whether its early formation contributes to the remaining unknown regions. To reduce unreliable control in the early high-noise stage, the trajectory maintains geometry guidance initially, gradually enables content signals, and limits the maximum deviation of local progress relative to overall progress. **Shuffle** maintains the same content distribution but swaps content information among mask samples of the same type

to break the correspondence between content and the current image.

Soft Phase and Content-v2 share a unified experimental protocol, with Content-v2 further integrating reliability and target utility information on top of geometry guidance.

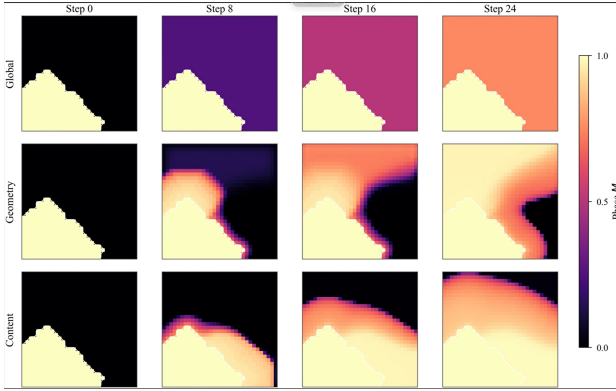


Fig. 1. Spatial generation progress under the same phase progression budget. Global synchronously advances all unknown regions, while Geometry and Content form non-uniform local progress distributions.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

First, mechanism positive controls are conducted on 32×32 RootedArrowCurve synthetic data. This data has a clear content-dependent formation order. A complete content-phase closed loop can recover the correct trajectory, while performance drops to random levels after shuffling or cutting off content feedback. This shows that the mechanism is learnable in data where content genuinely determines the generation order, but we cannot assume natural images follow the same rule.

Real image experiments use Places365 [12], with images processed to 256×256 and encoded into $4 \times 32 \times 32$ latents [13]. The experiments include four types of irregular masks, all retaining 25% known regions. Conditional images are masked in RGB space and encoded independently to prevent target information leakage. All strategies share a frozen latent velocity model, per-sample initial noise, and 32 main network calls. The primary metrics are unknown region latent MSE and known-unknown boundary latent MSE.

In the free generation trajectory audit, the median IQR of spatial precedence differences per image is 0, with only a few samples exhibiting significant local leading or lagging, thus not meeting the pre-specified criterion for endogenous asynchrony. As an exploratory analysis, a content probe trained on the target reference trajectory achieved an AUROC of 0.731, higher than Geometry’s 0.602 and Shuffle’s 0.597. This indicates that the current content in the target reference trajectory contains readable relative progress information, but this information does not naturally translate into stable control signals in free generation.

TABLE I
SPATIAL PHASE CONTROL RESULTS (LATENT MSE, LOWER IS BETTER)

Experimental Control Strategy	Unknown MSE ↓	Boundary MSE ↓
<i>Soft Phase</i>		
Global	0.971126	0.619622
Geometry	0.929540	0.472352
Content	0.962171	0.469221
<i>Content-v2</i>		
Reliability	0.961229	0.591209
Content-v2	0.954209	0.590267
Shuffle	0.956557	0.590334

B. Main Results

Geometry reduces the unknown region MSE by 4.28% and boundary MSE by 23.77% relative to Global, indicating that asynchronous progression relying solely on spatial geometry can yield stable benefits. Although the first version of Content achieved a slightly lower boundary error, its overall unknown region error was higher than Geometry, failing to establish an overall advantage.

Content-v2 improves unknown region MSE by 0.73% and 0.25% compared to Reliability and Shuffle, respectively. Paired bootstraps show consistent improvement directions across all four mask types. This demonstrates that correctly matched content information does provide a measurable incremental effect. However, both improvements fall short of the pre-specified 2% and 1% practical effect thresholds, and Content-v2 overall still does not surpass Geometry. Thus, current evidence supports that “content information has weak incremental value”, but it is insufficient to support it as the primary basis for spatial progression.

IV. DISCUSSION AND CONCLUSION

The results distinguish spatial asynchrony from content-driven control. Under the current latent inpainting setting, geometry-guided phase provides a stable spatial prior without increasing the number of main generative model calls. In contrast, although content-related progress information can be detected in target-reference trajectories, it does not yet provide sufficiently strong or reliable signals for closed-loop spatial control during free generation.

These findings suggest that readable content information is not necessarily equivalent to effective control information. The current study is limited to low-level latent representations and latent-domain reconstruction metrics, and therefore does not establish perceptual or state-of-the-art inpainting performance. Future work will investigate higher-level intermediate features, attention, and uncertainty representations from pretrained generative models, while further evaluating perceptual quality and computational efficiency on public inpainting benchmarks.

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